BioMaster Current Results Report

FDA-approved small molecules x human protein targets for cancer-oriented computational screening

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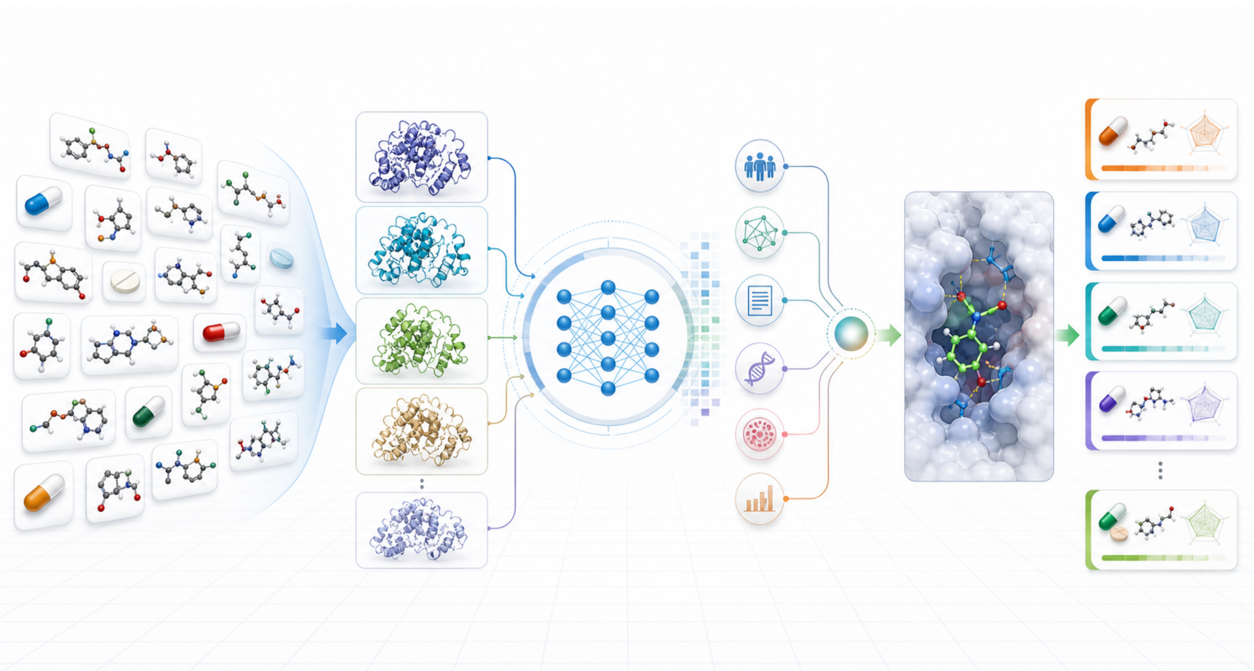

Figure 1. BioMaster screening workflow. Main figure generated with GPT-image-2. It illustrates the report-scale workflow from an FDA-approved drug library and human protein targets through AI-DTI screening, disease-evidence fusion, structural docking enhancement, and ranked candidate output. The figure intentionally contains no embedded text; quantitative labels and conclusions are provided in the LaTeX report.

Reporting position
The full Step 1-5 screening layer is complete. Stage 6 Top1000 structural enhancement has been produced. Full-scale DiffDock remains incomplete and should be described as a downstream structural-evidence expansion task.

1. Executive Summary

915	1000	915000	940/1000
FDA-approved small molecules	human protein targets	screened drug-target pairs	Top1000 DiffDock outputs

BioMaster has completed the Step 1-5 report-scale screening layer. The current run covers 915 FDA-approved small molecules and 1000 human protein targets, producing 915000 drug-target pairs. All pairs have ConPLex affinity or interaction predictions, followed by Open Targets and STRING disease-evidence ranking and TxGNN drug-disease evidence integration.

The Stage 5 Top1000 candidates have been structurally enhanced and merged into a Stage 6 consensus ranking. Among the 1000 candidate pairs, 940 have valid DiffDock outputs and 60 are missing or failed. The Stage 6 Top100 contains 67 unique drugs and 20 unique proteins, with 88/100 pairs having DiffDock outputs.

Critical boundary
Full-scale DiffDock is not complete. As of the 2026-05-11 audit, no full DiffDock queue process was detected. Out of 913170 DiffDock-ready pairs, 233500 have been processed, corresponding to 25.57 percent progress. Disk usage is 92.38 percent, so disk cleanup and a failed-pair rerun strategy are required before resuming the full queue.

2. Stage Completion

Stage	Status	Key result
Step 1 Drug library	Complete	915 FDA-approved small molecules. 915/915 have SMILES, InChIKey, molecular formula, and SDF. 913/915 have PubChem CID. DrugBank IDs are not yet integrated.
Step 2 Protein library	Complete	1000 proteins. 1000/1000 have sequences. 998/1000 have AlphaFold PDB/CIF receptor files. Missing receptors are P49895/DIO1 and P49908/SELENOP.
Step 3 ConPLex screening	Complete	915000 pairs have ConPLex predictions. No duplicate pairs and no missing manifest records were detected.
Step 4 AI ranking	Complete	915000 Stage 4 ranked pairs were generated. 913170 pairs are DiffDock-ready.
Step 5 Disease ranking	Complete	915000 pairs were ranked with Open Targets, STRING, and TxGNN evidence integration.
Step 6 Top1000 docking enhancement	Complete for Top1000	940/1000 pairs have DiffDock outputs. A Stage 6 consensus ranking has been produced.
Full DiffDock	Incomplete	233500/913170 pairs processed, 25.57 percent progress. The queue is currently not running.

3. Data and Evidence Integration

Source or model	Current integration	Reporting note
ChEMBL/ PubChem	915 drugs have ChEMBL IDs and structure fields. 913 have PubChem CIDs.	Sufficient for screening and structure preparation. DrugBank licensing and mapping remain follow-up items.
AlphaFold	998 proteins have local PDB/CIF receptor structures.	Two proteins lack usable AlphaFold paths and are excluded from DiffDock-ready pairs.
ConPLex	915000 drug-target pairs predicted.	Used as the scalable first-pass AI-DTI screening layer.
Open Targets	1987 target-disease evidence rows. Cancer direction matches 992 screening targets.	Primary disease ID is MONDO_0004992 / cancer.
STRING v12.0	2547 high-confidence human network edges.	Used for disease-related network evidence propagation.
TxGNN	738/915 drugs mapped. 738000 pairs have a TxGNN component.	Unmapped drugs retain Open Targets/STRING scores and are not treated as negative evidence.
DiffDock	Top1000 has 940 completed outputs. Full structural expansion is 25.57 percent complete.	DiffDock supports structural pose and confidence evidence. It is not a prerequisite for Step 1-5 completion.

4. Ranking Definitions

Stage 5 ranks drug-target pairs, not unique drugs. The Open Targets + STRING priority score is:

$$\text{OpenTargets\_STRING\_priority} = 0.55 \times \text{AI score} + 0.30 \times \text{OpenTargets} + 0.15 \times \text{STRING}$$

When TxGNN mapping is available, the Stage 5 final score is:

$$\text{final\_priority} = 0.80 \times \text{OpenTargets\_STRING\_priority} + 0.20 \times \text{TxGNN}$$

Stage 6 adds normalized DiffDock confidence:

$$\text{Stage6\_consensus} = 0.85 \times \text{Stage5\_final\_priority} + 0.15 \times \text{normalized\_DiffDock\_confidence}$$

5. Candidate Results

Stage 5 Top1000 covers 337 unique drugs and 64 unique proteins. All Top1000 pairs have direct and network disease evidence. The most frequent targets in the Top1000 are EGFR, TP53, BCL2, JAK1, TSHR, TACR1, GNRHR, and KIT.

5.1 Stage 5 Top 10 Unique-Drug Representative Pairs

Rank	Drug	Target	Final priority
1	Afatinib Dimaleate	EGFR	0.926857684
2	Pazopanib Hydrochloride	KIT	0.901545965
3	Dacomitinib	EGFR	0.889965753
4	Osimertinib Mesylate	EGFR	0.867670073
5	Cabozantinib S-Malate	KIT	0.864187850
6	Momelotinib Dihydrochloride	EGFR	0.861371239
7	Lazertinib Mesylate	EGFR	0.860511039
8	Repotrectinib	JAK1	0.860247310
9	Tucatinib	EGFR	0.845577840
10	Neratinib Maleate	EGFR	0.841781555

5.2 Stage 6 Consensus Top 10 Pairs

Rank	Drug	Target	Stage 5	Stage 6	Structural status
1	Afatinib Dimaleate	EGFR	0.926858	0.917979	completed
2	Dacomitinib	EGFR	0.889966	0.874261	completed
3	Cabozantinib S-Malate	KIT	0.864188	0.864188	missing output
4	Momelotinib Dihydrochloride	EGFR	0.861371	0.856885	completed
5	Repotrectinib	JAK1	0.860247	0.850873	completed
6	Pazopanib Hydrochloride	KIT	0.901546	0.848898	completed
7	Osimertinib Mesylate	EGFR	0.867670	0.839205	completed
8	Trilaciclib Dihydrochloride	JAK1	0.797277	0.827686	completed
9	Neratinib Maleate	EGFR	0.841782	0.808585	completed
10	Tucatinib	EGFR	0.845578	0.806943	completed

These results are computational priorities, not experimental proof of efficacy or mechanism. Experimental selection should still account for cancer subtype, novelty, drug accessibility, safety profile, known indication context, and assay feasibility.

6. Full DiffDock Status

Metric	Value
DiffDock-ready pairs	913170
Total jobs	3653
Generated score files	934
Processed pairs	233500
Pair progress	25.57%
Completed outputs	185696
Missing outputs	47804
Success rate among processed pairs	79.53%
Zero-completed chunks	64
Current queue process	Not detected
GPU usage	0 MB, 0% utilization
Disk usage	92.38%

The full DiffDock run should not be described as complete or actively stable. The old PID file still contains 178900, but process inspection found no active queue or job process. The practical next step is to free disk space and then decide whether to resume the full queue or run a narrower priority rerun.

7. Recommended External Wording

Recommended statement
We have completed full AI-DTI screening for 915 FDA-approved small molecules against 1000 human protein targets, covering 915000 drug-target pairs. The results integrate ConPLex predictions with Open Targets, STRING, and TxGNN evidence for a cancer-oriented disease ranking. The Stage 5 Top1000 candidates have been structurally enhanced with DiffDock, with 940/1000 valid outputs and a Stage 6 consensus ranking. Full-scale DiffDock is currently about 25.57 percent complete and should be treated as downstream structural-evidence expansion rather than a blocker for the Step 1-5 screening conclusion.

Avoid these claims
Do not say full DiffDock is complete. Do not interpret DiffDock missing outputs as non-binding evidence. Do not state that the computational results prove drug efficacy. Do not state that DrugBank mapping is complete.

8. Recommended Next Steps

1. Define the target disease scope more precisely. If moving from pan-cancer to a specific cancer type, rerun disease-evidence scoring with the appropriate disease ID.
2. Down-select 20-50 candidate pairs from the Stage 6 Top100 using literature support, novelty, safety, accessibility, and assay feasibility.
3. Prioritize reruns for the 12 missing-output pairs in Stage 6 Top100.
4. Free disk space before resuming full DiffDock; current disk usage is above a safe operating threshold.
5. Build a failed-pair audit and SMILES/parameter rerun queue for the 64 zero-completed chunks.

6. Add licensed DrugBank mapping later for indication, contraindication, and repositioning-compliance checks.

9. Verification

The repository tests were rerun before preparing this report:

```
pytest -q
8 passed in 0.54s
```

This report is based on the local 2026-05-11 audit snapshot and local result files. Large outputs under data, outputs, and third_party are local artifacts and are not part of the lightweight source repository release boundary.